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Data Science

PYTHON FILE HANDLING

1. Create and write to file

```
In [36]: with open("example.txt", "w") as file:  
         file.write("File created successfully!")
```

2. Read file

```
In [38]: with open("example.txt", "r") as file:  
         print(file.read())
```

File created successfully!

3. Overwrite file

```
In [40]: with open("example.txt", "w") as file:  
         file.write("Hello, this is Python file handling!")
```

4. Append new line

```
In [42]: with open("example.txt", "a") as file:  
         file.write("\nAdding new line.")
```

5. Read full content again

```
In [44]: with open("example.txt", "r") as file:  
         print(file.read())
```

Hello, this is Python file handling!
Adding new line.

6. Read line by line

```
In [46]: with open("example.txt", "r") as file:
        for line in file:
            print(line.strip())
```

Hello, this is Python file handling!
Adding new line.

7. Read only one line

```
In [48]: with open("example.txt", "r") as file:
        print(file.readline())
```

Hello, this is Python file handling!

8. Working with File Pointer

- Control where Python reads/writes in the file.

```
In [50]: with open("example.txt", "r") as file:
        print(file.read(5))    # read first 5 characters
        file.seek(0)          # move pointer back to start
        print(file.read())
```

Hello
Hello, this is Python file handling!
Adding new line.

9. Tell Current Position

```
In [54]: with open("example.txt", "r") as file:
        print(file.tell())    # position before reading
        file.read(10)
        print(file.tell())    # position after reading
```

0
10

10. Read Lines as List

```
In [57]: with open("example.txt", "r") as file:
        lines = file.readlines()
        print(lines)
```

['Hello, this is Python file handling!\n', 'Adding new line.']

11. Writing Multiple Lines

```
In [60]: lines = ["Line 1\n", "Line 2\n", "Line 3\n"]

with open("example.txt", "w") as file:
    file.writelines(lines)
```

12. File Copy

```
In [63]: with open("example.txt", "r") as src:
        content = src.read()

with open("copy.txt", "w") as dest:
    dest.write(content)
```

13. Rename File

```
In [67]: import os
os.rename("example.txt", "new_example.txt")
```

14. Delete File

```
In [71]: import os
os.remove("new_example.txt")
```

15. Working with CSV Files

```
In [74]: import csv

with open("data.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Name", "Age"])
    writer.writerow(["Riya", 21])
```

16. Working with JSON Files

```
In [77]: import json

data = {"name": "Riya", "age": 21}

with open("data.json", "w") as file:
    json.dump(data, file)

with open("data.json", "r") as file:
    print(json.load(file))
```

```
{'name': 'Riya', 'age': 21}
```

Things to remember :

- Always close files or use with open()
- "w" will erase existing content
- Use "a" if you don't want to lose old data